CLARA Monthly Tagup

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Planning and Plan Recognition

- **Planning:** from an initial state, find actions that achieve a goal
- **PDDL:** standard language for planning domains and problems
  - objects, predicates, actions with preconditions and effects
- **Action costs, goal utilities:** a plan's value is the utility of the goals it reaches minus the cost of its actions
- **Disjunctive goals:** the agent pursues *one* of several possible goals
- **Plan recognition** runs planning in reverse: observe some actions, infer which goal the agent is pursuing

Domain: Logistics

- Packages, trucks, and airplanes
  - trucks drive between places on roads
  - airplanes fly between airports
  - packages are loaded, unloaded, and transferred between vehicles
- Driving and flying have costs; loading is free
- Finding a minimum-cost plan is NP-complete
  - one truck delivering packages to *n* cities, with a cost on each road, is the traveling salesman problem

From Planning to Plan Recognition

- How do we get a probability distribution from a planning problem plus evidence?
- Treat every plan τ up to horizon *H* as possible, with probability exponential in its net utility:

$$P(\tau) = \frac{e^{U(\tau)}}{Z}, \qquad U(\tau) = \sum_{g \in \mathrm{goals}(\tau)} u_g \;-\; \sum_{a \in \tau} c_a$$

- Probability of goal *g* given observations *O* ($\tau \models O \wedge g$: plan τ fits *O* and reaches *g*):

$$P(g \mid O) = \frac{\sum_{\tau \models O \wedge g} e^{U(\tau)}}{\sum_{\tau \models O} e^{U(\tau)}}$$

Example



Each move costs 1 · each goal is worth +4 · horizon 3 (waiting is free)

Plans ending at	No evidence	After a move right
g1	**0.72** (3 plans of U = 3, 2 of U = 1)	**0.15** (1 plan of U = 1)
g2	**0.24** (3 plans of U = 2)	**0.81** (2 plans of U = 2)
neither	0.03 (13 plans)	0.04 (5 plans)

g1 is still possible after moving right, but much less likely

Baseline: Translation to FiFO

- **FiFO:** finite-domain first-order logic, compiled to propositional CNF
- **Weighted logic (Markov logic):** each weighted formula adds its cost when true

$$P(x) \propto \exp\Big(-\sum_{i \,:\, x \models f_i} w_i\Big)$$

- **SATPLAN:** a PDDL problem at horizon *H* becomes CNF over atoms *holds(f, t)* and *occurs(a, t)*
  - each satisfying assignment is a plan; action costs become weights
- Plan recognition is then a ratio of weighted model counts:

$$P(g \mid O) = \frac{\mathrm{WMC}(T \wedge O \wedge g)}{\mathrm{WMC}(T \wedge O)}$$

Model Counting

- **Exact: SharpSAT-TD**
  - DPLL search with component caching, guided by a tree decomposition
  - exact answers; #P-hard in general
- **Approximate: Monte Carlo (MC-SAT)**
  - Markov chain whose steps sample satisfying assignments with a SAT-based sampler
  - all marginals from one run; can mix slowly

Max-Model Approximation

- Replace each sum over plans by its largest term: the cheapest plan (Ramírez & Geffner 2010)
- For each goal *g*, two MaxSAT calls:
  - c(g, O): cheapest plan to *g* consistent with the observations
  - c(g, ¬O): cheapest plan to *g* that is not

$$P(g \mid O) \propto \sigma\big(\beta\,(c(g,\neg O) - c(g, O))\big)$$

- No counting, only optimization: 2*n* MaxSAT calls for *n* goals
- Ignores *how many* plans there are, so goals with equally cheap plans tie

Results on Simple Grids

Grid	Goals	CNF vars	CNF clauses	Max-term P / time	SharpSAT-TD P / time	MC-SAT P / time
3×3	2	141	818	0.50 / 12 s	0.50 / 16 s	0.51 / 14 s
4×4	3	400	2,609	0.33 / 16 s	0.37 / 20 s	0.36 / 33 s
5×5	4	812	5,621	0.26 / 35 s	0.27 / 35 s	0.25 / 210 s
6×6	5	1,557	11,191	0.20 / 30 s	0.21 / 38 s	0.19 / 349 s
7×7	6	2,436	17,936	0.18 / 40 s	0.20 / 105 s †	0.19 / 737 s †
8×8	7	4,024	30,205	0.15 / 61 s	none finished	none finished

- P = mean probability given to the true goal (chance = 1 / goals); time = mean per run
- Counting does not scale: † on 7×7, 36 SharpSAT-TD and 2 MC-SAT runs of 144 hit the 20-minute limit; none finished on 8×8
- Max-term always finishes but ties every goal: early moves on an open grid say little about the goal

Knowledge Compilation

- Compile the problem once to a d-DNNF circuit (d4), with no evidence
- Then marginals given *any* observed moves come from one pass over the circuit

Grid	Horizon	CNF vars	CNF clauses	Compile time	Circuit nodes	Query, 2 obs.
3×3	5	141	818	0.04 s	44	0.001 s
5×5	8.5	812	5,621	0.09 s	1,108	0.007 s
6×6	10.75	1,557	11,191	1.4 s	21,140	0.017 s
7×7	12	2,436	17,936	6.8 s	82,748	0.035 s
8×8	14.75	4,024	30,205	104 s	785,660	0.24 s
9×9	16	5,616	42,712	418 s	2,525,681	0.45 s

10×10 did not compile within 20 minutes

Results on Two-Package Grids

One truck, two packages, two possible destinations each; the agent delivers one package: 4 goals (chance = 0.25), 4 instances per grid, 1 … N−2 observed actions

Grid	CNF vars	CNF clauses	Max-term P / time	SharpSAT-TD P / time	MC-SAT P / time	d4 compile done · time
3×3	751	10,533	0.40 / 24 s	0.46 / 28 s	0.37 / 85 s	4 of 4 · 4 s
4×4	1,893	40,065	0.35 / 35 s	0.39 / 52 s	0.31 / 454 s	4 of 4 · 5 s
5×5	3,147	93,878	0.39 / 56 s	0.41 / 67 s	0.32 / 579 s †	4 of 4 · 9 s
6×6	5,797	234,150	0.39 / 103 s	none finished	not run	4 of 4 · 105 s
7×7	9,925	523,734	0.39 / 355 s	not run	not run	2 of 4 · 207 s

- P = mean probability given to the true goal over finished runs; time = mean per run; CNF sizes are means over the 4 problems; 20-minute limit; a method that timed out was not run on larger grids
- Counting stops early: SharpSAT-TD finished none of 8 runs on 6×6; † MC-SAT finished 4 of 16 on 5×5
- Max-term finishes everything, within 0.06 of exact counting, but its top goal is a tie in 216 of 240 runs
- d4 compiles through 6×6; each query then takes ≤ 0.26 s and matches SharpSAT-TD to $2 \cdot 10^{-15}$

Test Problem Archive

- **GitHub:** github.com/HenryKautz/CLARA
  - tools: github.com/HenryKautz/FiFO
- **Problem generator:** ppgen makes problems, planner finds plans, evgen turns plans into observations
- **Simple grid instances:** N×N grids, N = 3 … 10, one truck, several possible destinations
- **Two-package instances:** N×N grids, N = 3 … 7, two packages with two possible destinations each
- **Baseline results:** the tables on the previous three slides